

Amplitude amplification

Amplitude amplification is a technique in quantum computing which generalizes the idea behind Grover's search algorithm, and gives rise to a family of quantum algorithms. It was discovered by Gilles Brassard and Peter Høyer in 1997,^[1] and independently rediscovered by Lov Grover in 1998.^[2]

In a quantum computer, amplitude amplification can be used to obtain a quadratic speedup over several classical algorithms.

Algorithm

The derivation presented here roughly follows the one given by Brassard et al. in 2000.^[3] Assume we have an N -dimensional Hilbert space $\mathcal{H}$ representing the state space of a quantum system, spanned by the orthonormal computational basis states $B := \{|k\rangle\}_{k=0}^{N-1}$. Furthermore assume we have a Hermitian projection operator $P: \mathcal{H} \rightarrow \mathcal{H}$. Alternatively, P may be given in terms of a Boolean oracle function $\chi: \mathbb{Z} \rightarrow \{0, 1\}$ and an orthonormal operational basis $B_{\text{op}} := \{|\omega_k\rangle\}_{k=0}^{N-1}$, in which case

$$P := \sum_{\chi(k)=1} |\omega_k\rangle\langle\omega_k|.$$

P can be used to partition $\mathcal{H}$ into a direct sum of two mutually orthogonal subspaces, the *good* subspace $\mathcal{H}_1$ and the *bad* subspace $\mathcal{H}_0$:

$$\begin{aligned}\mathcal{H}_1 &:= \text{Image } P &= \text{span}\{|\omega_k\rangle \in B_{\text{op}} \mid \chi(k) = 1\}, \\ \mathcal{H}_0 &:= \text{Ker } P &= \text{span}\{|\omega_k\rangle \in B_{\text{op}} \mid \chi(k) = 0\}.\end{aligned}$$

In other words, we are defining a "*good subspace*" $\mathcal{H}_1$ via the projector P . The goal of the algorithm is then to evolve some initial state $|\psi\rangle \in \mathcal{H}$ into a state belonging to $\mathcal{H}_1$.

Given a normalized state vector $|\psi\rangle \in \mathcal{H}$ with nonzero overlap with both subspaces, we can uniquely decompose it as

$$|\psi\rangle = \sin(\theta)|\psi_1\rangle + \cos(\theta)|\psi_0\rangle,$$

where $\theta = \arcsin(|P|\psi\rangle|) \in [0, \pi/2]$, and $|\psi_1\rangle$ and $|\psi_0\rangle$ are the normalized projections of $|\psi\rangle$ into the subspaces $\mathcal{H}_1$ and $\mathcal{H}_0$, respectively. This decomposition defines a two-dimensional subspace $\mathcal{H}_\psi$, spanned by the vectors $|\psi_0\rangle$ and $|\psi_1\rangle$. The probability of finding the system in a *good* state when measured is $\sin^2(\theta)$.

Define a unitary operator $Q(\psi, P) := -S_\psi S_P$, where

$$\begin{aligned}S_\psi &= I - 2|\psi\rangle\langle\psi| \quad \text{and} \\ S_P &= I - 2P.\end{aligned}$$

S_P flips the phase of the states in the *good* subspace, whereas S_ψ flips the phase of the initial state $|\psi\rangle$.

The action of this operator on $\mathcal{H}_\psi$ is given by

$$\begin{aligned}Q|\psi_0\rangle &= -S_\psi|\psi_0\rangle = (2\cos^2(\theta) - 1)|\psi_0\rangle + 2\sin(\theta)\cos(\theta)|\psi_1\rangle \text{ and} \\ Q|\psi_1\rangle &= S_\psi|\psi_1\rangle = -2\sin(\theta)\cos(\theta)|\psi_0\rangle + (1 - 2\sin^2(\theta))|\psi_1\rangle.\end{aligned}$$

Thus in the $\mathcal{H}_\psi$ subspace Q corresponds to a rotation by the angle 2θ :

$$Q = \begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ \sin(2\theta) & \cos(2\theta) \end{pmatrix}.$$

Applying Q n times on the state $|\psi\rangle$ gives

$$Q^n|\psi\rangle = \cos((2n+1)\theta)|\psi_0\rangle + \sin((2n+1)\theta)|\psi_1\rangle,$$

rotating the state between the *good* and *bad* subspaces. After n iterations the probability of finding the system in a *good* state is $\sin^2((2n + 1)\theta)$.

The probability is maximized if we choose

$$n = \left\lfloor \frac{\pi}{4\theta} \right\rfloor.$$

Up until this point each iteration increases the amplitude of the *good* states, hence the name of the technique.

Applications

Assume we have an unsorted database with N elements, and an oracle function χ which can recognize the *good* entries we are searching for, and $B_{\text{op}} = B$ for simplicity.

If there are G *good* entries in the database in total, then we can find them by initializing a quantum register $|\psi\rangle$ with n qubits where $2^n = N$ into a uniform superposition of all the database elements N such that

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle$$

and running the above algorithm. In this case the overlap of the initial state with the *good* subspace is equal to the square root of the frequency of the *good* entries in the database, $\sin(\theta) = |P|\psi\rangle| = \sqrt{G/N}$. If $\sin(\theta) \ll 1$, we can approximate the number of required iterations as

$$n = \left\lfloor \frac{\pi}{4\theta} \right\rfloor \approx \left\lfloor \frac{\pi}{4\sin(\theta)} \right\rfloor = \left\lfloor \frac{\pi}{4} \sqrt{\frac{N}{G}} \right\rfloor = O(\sqrt{N}).$$

Measuring the state will now give one of the *good* entries with high probability. Since each application of S_P requires a single oracle query (assuming that the oracle is implemented as a quantum gate), we can find a *good* entry with just $O(\sqrt{N})$ oracle queries, thus obtaining a quadratic speedup over the best possible classical algorithm. (The classical method for searching the database would be to perform the query for every $e \in \{0, 1, \dots, N-1\}$ until a solution is found, thus costing $O(N)$ queries.) Moreover, we can find all G solutions using $O(\sqrt{GN})$ queries.

If we set the size of the set G to one, the above scenario essentially reduces to the original Grover search.

Quantum counting

Suppose that the number of *good* entries is unknown. We aim to estimate $\tilde{G}$ such that $(1 - \delta)G \leq \tilde{G} \leq (1 + \delta)G$ for small $\delta > 0$. We can solve for $\tilde{G}$ by applying the quantum phase estimation algorithm on unitary operator Q .

Since $e^{2i\theta}$ and $e^{-2i\theta}$ are the only two eigenvalues of Q , we can let their corresponding eigenvectors be $|\psi_1\rangle$ and $|\psi_2\rangle$. We can find the eigenvalue $e^{2i\theta}$ of $|\psi\rangle$, which in this case is equivalent to estimating the phase θ . This can be done by applying Fourier transforms and controlled unitary operations, as described in the quantum phase estimation algorithm. With the estimate $\tilde{\theta}$, we can estimate $\sin \theta$, which in turn estimates G .

Suppose we want to estimate θ with arbitrary starting state $|s\rangle$, instead of the eigenvectors $|\psi_1\rangle$ and $|\psi_2\rangle$. We can do this by decomposing $|s\rangle$ into a linear combination of $|\psi_1\rangle$ and $|\psi_2\rangle$, and then applying the phase estimation algorithm.

References

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